EXAM PREPARATION

1. **Heart of the heart is:**
   a) SA Node
   b) AV Node
   c) Bundle of HIS
   d) Purkinji Fiber

2. **Which of the following has the thickest wall?**
   a) Right ventricle
   b) Left ventricle
   c) Right atrium
   d) Left atrium

3. Pacemaker is _____
a) AV node
b) SA node
c) Bundle of HIS
d) Ventricle muscles

4. SA node is located in _____
a) upper lateral wall of right atrium
b) lower lateral wall of left atrium
c) lower lateral wall of right atrium
d) upper lateral wall of left atrium

5. "Bundle of HIS" is made up of _____
a) nervous tissue supplied to ventricles
b) nervous tissue supplied to heart
c) muscular tissue supplied to heart
d) muscular tissue supplied to ventricles

6. Blood enters into the heart because of muscles of _____

a) atria relax

b) ventricles contract

c) ventricles relax

d) atria contract

7. The opening of right atrium into right ventricle is guarded by _____

a) mitral valve

b) tricuspid valve

c) bicuspid valve

d) aortic semilunar valve

8. Mitral valve is present between _____

a) right atrium and left ventricle

b) right and left ventricle

c) left ventricle and aorta

d) left atrium and left ventricle

9.Cardiac output is determined by _____

a) heart rate

b) stroke volume

c) blood flow

d) heart rate and stroke volume

10. In human being the duration of cardiac cycle is _____

a) 0.008 sec

b) 0.5 sec

c) 0.8 sec

d) 8 sec

11. The volume of blood each ventricle pumps out during a cardiac cycle is about _____

a) 70 ml

b) 5000 ml

c) 7 L

d) 1200 ml

12. What is the definition of the cardiac cycle?

a) The contraction of the atria

b) Circulation of the blood in the heart

c) The contraction and relaxation of the ventricles

d) It is a sequence of event that occurs during one complete heartbeat

13. Name the three phases of cardiac cycle in the order _____

a) early diastole, mid-to-late diastole, ventricular systole

b) ventricular systole, mid-to-late diastole, Early diastole

c) mid-to-late diastole, Ventricular systole, Early diastole

d) early diastole, Ventricular systole, mid-to-late diastole

14. An electrocardiogram is a graphic illustration of _____

a) cardiac conduction system

b) cardiac cycle

c) cardiac output

d) systemic and pulmonary circuits

15. The P wave of the ECG represents _____

a) ventricular depolarization

b) atrial depolarization

c) ventricular repolarization

d) atrial systole

16. Which of the following is not true for ventricular systole?

a) The ventricles relax

b) The ventricles contract

c) The semi lunar valves close

d) The atrioventricular valves open

17. The two distinct heart sounds during the cardiac cycle are lub and dup.

a) True

b) False

18. Heart beat initiates from _____

a) Purkinji fibers

b) SA node

c) Bundle of HIS

d) Auriculo ventricular node

19. An artificial pace maker is implanted and connected to the heart in patients _____

a) suffering from arteriosclerosis

b) having high blood pressure

c) with irregularity in heart rhythm

d) having 90% blockage

20. A heat murmurs indicates a defective _____

a) Heart valves

b) Bundle of HIS

c) SA node

d) AV node

21. Choose the correct pathway of the transmission of impulses in heartbeat.

a) AV node- SA node- Bundle of HIS- Purkinji fibres

b) SA node- AV node- Bundle of HIS- Purkinji fibres

c) SA node- Bundle of HIS- AV node- Purkinji fibres

d) SA node Purkinji fibres- Bundle of HIS- AV node

22. In ECG the depolarization of atria is indicated by _____

a) P wave

b) Q wave

c) R wave

d) S wave

23. Which of the following pair is correctly matched?

a) Dup- Sudden opening of semilunar valves at the beginning of ventricle diastole

b) Lubb- Sharp closing of AV valves at the beginning of ventricular systole

c) Pulsation of radial artery valves in blood vessels

d) Initiation of heart beat- Purkinji fibers

24. Which instrument is used to listen to the internal sounds of the human body?

a) Stethoscope

b) Sphygmomanometer

c) Reflex hammer

d) Suction device

25. The following are the major functional physiological systems of the body except?

A. Cardiovascular system

B. Respiratory system

C. Electrocardiogram system

D. Nervous system

26. Physiological Signals are generated by the body during the functioning of various physiological systems. Therefore, physiological signals hold information which can be extracted from these signals to find out the state of functioning of these physiological systems, true or false?

A. True

B. False

27. Which one of the following is not a transducer?
A. Thermocouple
B. Photovoltaic
C. Electrooculogram
D. Moving coil generator

28. Electroencephalogram is obtained from bioelectrical signals from the?
A. Brain
B. Heart
C. Muscles
D. Retina

29. The following are resistive passive transducers, except?
A. Strain gauge
B. Potentiometer
C. Linear variable differential transformer (LVDT)
D. Photodiode

30. The capacitance of a passive capacitance transducer depends on the following factors, except?
A. Distance between the two parallel plates
B. The area of the two parallel plates
C. Relative dielectric constant
D. Mass of the two parallel plates

31. Biopotentials are ionic voltages produced when certain types of cells carry out electrochemical activities, true or false?
A. True
B. False

32. The biopotentials generated by the muscles of the heart with time is called?
A. Electroretinogram
B. Electroencephalogram
C. Electrocardiogram
D. Electrooculogram

32. All the following are examples of biopotentials, true or false?

- Electroencephalogram (EEG)
- Electromyogram (EMG)
- Electroretinogram (ERG)

A. True
B. False

33. How can you define electromyogram (EMG)?
A. The recorded representation of bioelectric potentials generated by the activity of the brain (Neuronal activity.
B. The biopotentials generated by the muscles of the heart with time.
C. The recorded representation of bioelectric potentials generated by the muscle activity.
D. The measure of the variations in the corneal-retinal potential.

34. Electroencephalogram (EEG) is used in the diagnosis of the:
A. Heart
B. Tumour
C. Neuromuscular diseases
D. Brain

35. Which of the following statements is true about Needle electrodes?
A. They are designed to measure bioelectric potentials near or within the cell.
B. They are designed to penetrate the skin so that they can record biopotentials like EEG signals from the brain.
C. They are designed to measure biopotentials from the surface of the skin.
D. Typical examples include the metal microelectrodes and micropipette.

36. The following are statements about heart sounds:

- **The beating of the heart and the pumping of the blood is associated with the generation of the sounds.**
- **The technique of listening to sounds by the heart and blood vessels is called auscultation.**
- **The physicians are trained to diagnose the heart disorders by listening to these sounds using a device called the stethoscope.**
- **A graphic record of the heart is called phonocardiogram.**

Are all the above statements are true or false?
A. True
B . False

37. The following are methods of blood flow measurement, except?
A. Magnetic blood flow measurement
B. Coriolis blood flow measurement
C. Ultrasonic blood flow measurement
D. Radiographic blood flow measurement

38. The following procedure refers a blood flow measurement method:

- **The dye is injected at a constant rate in the blood flow and a detector measures the dye concentration downstream.**
- **The concentration of the dye in the blood stream keep on increasing at lapses and it finally reaches a constant value.**
- **The blood flow = Injection of dye (milligrams/mm)/concentration of dye (milligrams per liter)**

Which blood flow measurement method is described by the above procedure?
A. Plethysmography chamber blood flow meter
B. Radiographic flow meter
C. Dye dilution flow meter
D. Ultrasonic blood flow meter

39. All the following are methods of blood pressure measurement, except?
A. Sphygmomanometer
B. Percutaneous method
C. Hagen-poiseuille analysis
D. Catheterization

41. Electrocardiograph (ECG) is a graphic recording or display of the time variant produced by the heart during the cardiac cycle, true or false?
A. True
B. False

42. What is Tidal volume with regard to lung capacities?
A. It is the extra volume that can be inspired.

B. It is the maximum volume of the gas that can be expelled from the lungs by forceful expiration after maximum inspiration.
C. It is the volume of gas inspired or expired during each respiration cycle.
D. It is the volume of air remaining in the lungs at the end of expiratory level.

43. What biological measurement is done by the Spirometer?
A. Blood pressure measurement
B. Blood Flow measurement
C. Respiratory volume measurements
D. Blood sugar measurement

44. What is the function of a Nebulizer as a respiratory therapy?
A. It is a device used to administer medication in the form of mist inhaled into the lungs and used for treatment of asthma, and other pulmonary disorders.
B. It is a device used to remove liquid or gases by suction from the body.
C. It is equipment used to increase humidity of the inspired air.
D. None of the above.

45. The following are properties of ultrasound waves as applied in medical imaging, except?
A. They are longitudinal
B. They are acoustic
C. They are electromagnetic
D. They depend upon the medium through which it propagates

46. Source of Bioelectric potential is _____ in nature.
a) electronic
b) electric
c) ionic
d) mechanical

47. Palsied muscles mean _____
a) paralyzed muscles
b) active muscles
c) voluntary muscles
d) involuntary muscles

48. The principal ion that is not involved with the phenomena of producing cell potentials is _____
a) sodium
b) potassium
c) chlorine
d) hydrogen

49. What is the relatively static membrane potential of quiescent cells called?
a) half-cell potential
b) action potential
c) resting membrane potential
d) cell potential

50. The variation of the electrical potential associated with the passage of a pulse along the membrane of a muscle cell or a nerve cell is called _____
a) muscle potential
b) action potential
c) resting potential
d) half-cell potential

51. Cells depolarize and action potential in generated as soon as a stimulus is applied.
a) True
b) False

52. After a cell is stimulated, a finite period of time is required for the cell to return to its pre-stimulus state. This period is known as _____
a) restoration period
b) refactory period
c) regain period
d) regenerative period

53. Electrooculography (EOG/E.O.G.) is a technique for measuring what?
a) abnormal function of the retina
b) heart rate
c) respiration rate
d) cornea-retinal standing potential

54. EKG stands for _____
a) Electrocardiography

b) Electroencephalography
c) Electromyography
d) Electrtokinetcography

55. Phonocardiography is listening to _____
a) arm muscle sound
b) lungs sound
c) heart sound
d) respiratory tract sound

56. Which of the following is a preferred electrode for measuring EMG?
a) surface electrodes
b) needle electrodes
c) pregelled electrodes
d) scalp electrodes

57. Generally what is the material of needle electrodes?
a) stainless steel
b) copper
c) lead
d) iron

58. Monopolar needle electrodes are having coatings of which material over the stainless steel wires which are bare only at the tips?
a) carbon
b) calcium
c) sodium
d) teflon

59. Which electrode can be used to pick up signals from individual fibers of muscle tissues?
a) biopolar needle electrode
b) concentric core needle electrode
c) multi-element needle electrode
d) monopolar needle electrode

60. _____ instrument is used to hold patients head and guide the placement of electrodes.
a) Monotaxic
b) Stereotonic
c) Stereotaxic

d) Monotonic

61. The ground electrode is usually positioned over which body structures?
a) bony
b) hairy
c) fleshy
d) sweaty

62. When intramuscular EMG is required to look into the electrical activities of deeper or overlaid muscles, _____ electrodes are used.
a) plate shape electrodes
b) surface electrodes
c) thin thread electrodes
d) fine wire electrodes

63. The contraction of the skeletal muscles results in the generation of action potential in the individual muscle fibers. Record of this action potential is called _____
a) ECG
b) EMG
c) EEG
d) EKG

64. In voluntary contraction of the skeletal muscles, the muscle potential ranges from _____
a) 50 uV – 5 mV
b) 50 mV – 5 V
c) 0.05 uV – 2 mV
d) 50 mV – 500 mV

65. Electrodes to measure EEG are placed on _____
a) forehead
b) scalp
c) cheek
d) ears

66. According to the international 10/20 system to measure EEG, even number denotes which side of the brain?
a) left
b) top
c) bottom

d) right

67. Letter F in the EEG electrode placement system denotes?
a) front
b) face
c) frontal lobe
d) fast

68. Normal EEG frequency range is _____
a) 50-500Hz
b) 0.5-50HZ
c) 0.05-5Hz
d) 1-200Hz

69. The letter T in the EEG electrode placement system denotes?
a) temporal lope
b) temper lobe
c) trace
d) timpanic

70. According to the international 10/20 system to measure EEG, odd number denotes which side of the brain?
a) left
b) right
c) top
d) front

71. The delta wave in EEG ranges from _____
a) 0.5-4Hz
b) 4-8Hz
c) 8-13Hz
d) 13-22Hz

72. Disturbance in the EEG pattern resulting from the external stimuli is called _____
a) provoked response
b) ckoored response
c) evoked response
d) impulse response

73. The peak to peak amplitude of the waves that can be picked from the scalp is _____

a) 100mV
b) 100V
c) 100uV
d) 10mV

74. Which rhythm is the principal component of the EEG that indicates the alertness of the brain?

a) theta rhythm
b) gamma rhythm
c) beta rhythm
d) alpha rhythm